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Plenary Lectures



The Violent Universe: A Multi-Messenger View

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Abstract

The Universe is filled with extreme and violent phenomena: exploding stars, powerful jets from black holes, neutron-star mergers, and other energetic cosmic events. To understand these phenomena, astronomers can no longer rely on light alone. Today, we explore the Universe through several different cosmic messengers, each revealing a different aspect of the most energetic events in nature.

This multi-messenger approach combines **gamma rays**, **cosmic rays**, **neutrinos**, and **gravitational waves**. Gamma rays trace some of the most powerful electromagnetic processes, while cosmic rays carry information about the energetic particles that travel across the Universe. Neutrinos can escape dense and distant astrophysical environments and provide unique clues about cosmic particle acceleration, while gravitational waves allow us to observe the dynamics of merging compact objects through ripples in spacetime.

Together, these messengers provide complementary views of the same cosmic sources and open a new way of investigating the extreme Universe. In this talk, we will discover what each messenger can tell us, how they are detected, and how combining them is transforming our understanding of the violent and fascinating Universe.

Keywords: *Multi-messenger astronomy; Gamma rays; Cosmic rays; Neutrinos.*



Black Holes and Active Galactic Nuclei: An X-Ray Perspective

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Abstract

It is now widely accepted that supermassive black holes lie at the heart of most galaxies. A subset of these black holes actively accretes matter, powering Active Galactic Nuclei (AGN). Within these systems, interactions between the accretion disk and the hot corona generate intense X-ray radiation.

Consequently, X-ray spectroscopy and timing serve as essential tools to analyze the structural components of AGN and constrain their physical properties. In this presentation, we review how X-ray observations allow us to probe the inner workings of AGN and map the physical properties of these powerful N and map the physical properties of these powerful cosmic drivers.



Thematic Lectures



Technological Watch and AI Innovations in Space Applications: Key Research Activities at the Algerian Space Agency

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Abstract

This presentation provides a comprehensive overview of Algeria's space initiatives and the integration of emerging technologies within national space programs, structured into two complementary sections:

The first part introduces the Algerian Space Agency (ASAL), detailing its strategic mandate, organizational infrastructure across specialized operational centers, and space assets—most notably the Alsat Earth observation constellation and the Alcomsat-1 telecommunications satellite. It highlights operational downstream applications serving national socio-economic priorities, including natural risk management, agriculture, water resource monitoring, and regional land use planning.

The second part focuses on technological watch and artificial intelligence (AI) innovations within ASAL. It explores how continuous scientific monitoring guides the adoption of state-of-the-art machine learning paradigms in space applications. Key research activities are highlighted. Finally, the presentation discusses operationalization, academic collaborations, and future perspectives for developing robust Geo-AI solutions tailored to sustainable national development



Observing the Sky with a Small Telescope

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Abstract

Engaging with observational astronomy through optical instruments has profoundly shaped my academic trajectory. During my doctoral research at the Haute-Provence Observatory, I had the privilege of operating telescopes ranging from 80cm and 120cm apertures to larger 193cm and 152cm facilities.

A primary objective of my current work is to bridge the gap between professional instrumentation and youth education in my home country. Providing students with direct access to optical astronomy fosters critical thinking and expands scientific literacy among the next generation.

Presentation Overview:

- **Instrument Selection:** Guidance on choosing an appropriate small- aperture telescope tailored to specific observational needs.
- **Optics and Components:** A technical breakdown of telescope anatomy and foundational operating principles.
- **Performance Limitations:** Realistic expectations regarding atmospheric conditions, resolution limits, and optical aberrations.
- **Session Planning:** Methodologies for structuring and optimizing observing sessions to maximize scientific yield and visual experience.



Muonic Tomography: Considerations on a Fascinating Technique

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Abstract

The muon, an atypical elementary particle, was discovered in 1936 by American physicists Carl D. Anderson and Seth Neddermeyer [1]. Muons are generated by the interactions between cosmic rays (astroparticles, made up of protons and helium nuclei) and atoms in Earth's upper atmosphere. The latter includes regions between 20 and 100 km (upper stratosphere, mesosphere and thermosphere). With a lifetime of 2.2 μ s, they were supposed to disintegrate before reaching the ground. Einstein's relativistic theories (time dilation, length contraction) explain the opposite fact.

This allowed, thanks to the genius of Physicists, the development of Muography or, more academically, Muonic Tomography. This fascinating technique belongs to the techniques of Imaging and is based on the attenuation or diffusion of muons during their crossing of a massive and opaque object [2].

The technique has been consecrated as powerful in Archaeology, Volcanology, Nuclear Industry and Security [3]. In this talk, we will present some considerations on a prospective implementation of this technique, particularly in the studies of our vast cultural heritage (Royal Mausoleum of Mauretania, Madghacen, Ledjars). Inspired by the ScanPyramids project [4], one could only garner beneficial know-how, given the local manpower and its qualifications. Opportunities for joint ventures with cultural administrations responsible for human heritage and local communities will also be discussed.

Keywords: *Muon, Tomographie Muonique, Héritage culturel.*

References :

[1] C. Benchouk, *Low-Cost high-energy Physics (JSP'17, USTHB, 23-24 avril 2013)*.

[2] P. Checchia, *Review of possible applications of cosmic muon tomography, 2016 JINST 11 C12072*.

[3] C. Benchouk-D. Bradai, *Private discussions (June, 2026)*



Numerical Simulation of Meteoroid Atmospheric Entry and Extraction of the Associated Pressure Signature

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Abstract

Meteoroid atmospheric entry is a highly energetic hypersonic phenomenon involving the formation of strong shock waves, intense aerodynamic heating, and significant pressure disturbances in the surrounding atmosphere. In this work, a simplified numerical simulation of meteoroid atmospheric entry is performed using computational fluid dynamics. A spherical body is considered under representative upper atmospheric conditions and hypersonic entry velocity. The numerical model is used to investigate the bow shock structure and the distributions of pressure, temperature, density, and flow velocity around the body. Particular attention is given to the pressure disturbance generated by the hypersonic flow. Pressure profiles are extracted at different locations relative to the meteoroid trajectory in order to examine the amplitude and spatial evolution of the generated disturbance. The study aims to provide a simplified physical description of the relationship between meteoroid hypersonic entry, shock wave formation, and the resulting pressure signature. Such numerical investigations may contribute to a better understanding of the physical source mechanisms associated with acoustic and infrasonic signals generated by meteoroid atmospheric entries.

Keywords: Hypersonic flow, Meteoroid, Acoustic waves, Atmospheric entry, Computational Fluid Dynamics.



Fundamentals of Space Systems: Satellite Technology, Mission Design, Remote Sensing, and Satellite Subsystems.

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Abstract

Space systems play a key role in addressing a wide range of scientific, technological, environmental, and communication needs. This presentation provides an overview of the fundamental concepts underlying satellite technology and space mission design, with particular emphasis on the architecture, technologies, and applications of modern space systems. The presentation introduces the main components of a space system, including the space segment, the ground segment, and the communication link between the satellite and ground infrastructure. The mission analysis process is then discussed through the identification of mission needs, objectives, and the selection of appropriate satellite and mission concepts.

Particular attention is given to Earth observation missions and their associated remote sensing technologies. Different types of on-board instruments, including optical, multispectral and hyperspectral sensors, synthetic aperture radar (SAR), radiometers, and altimeters, are introduced in relation to their applications in mapping, natural resource management, environmental monitoring, and climate observation. The presentation also addresses satellite telecommunication missions, including their payloads, frequency bands, antennas, and ground infrastructure.

Finally, the major environmental constraints encountered during launch and in orbit are presented, including mechanical loads, vibrations, acoustic effects, thermal conditions, radiation, vacuum and material outgassing. Examples from Algerian space activities and satellite missions, including the ALSAT series and ALCOMSAT-1, are used to illustrate the progressive appropriation of space technologies and the development of national capabilities.

Keywords: *Satellite Technology, Remote Sensing, Satellite Subsystems, Earth Observation, Space Missions.*



Introduction to Observation in Cosmology

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Abstract

This presentation explains how we learn about the universe by observing it. The discussion covers why looking at things is so important for understanding how the universe works. The presentation explores different ways to observe and gather information, and why using multiple methods helps us understand the bigger picture. The discourse addresses major questions about what exists in the universe and what might happen to it in the future. The presentation concludes by discussing new methods and tools that will help us observe and learn more about space. Attendees will understand how observing things around us helps scientists answer big questions about the universe.



From the Big Bang to the Accelerating Universe: A Journey Through Modern Cosmology

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Abstract

Modern cosmology seeks to understand the origin, evolution, and large-scale structure of the Universe. Over the past century, observations and theoretical developments have transformed our view of the cosmos, from the discovery of cosmic expansion to the establishment of the standard Λ CDM model. This presentation provides a broad overview of the main ideas and observational evidence underlying our current understanding of the Universe.

We begin with the expanding Universe and the Big Bang framework, discussing key observational evidence such as the Hubble–Lemaître law, the cosmic microwave background, and the primordial abundance of light elements. We then examine some of the fundamental challenges of the standard Big Bang model, including the horizon and flatness problems, which motivate the inflationary scenario. The role of inflation in the early Universe and the generation of primordial fluctuations that later seeded cosmic structures will be briefly discussed.

The presentation then turns to the dark sector of the Universe, focusing on the role of dark matter in structure formation and the evidence for the late-time accelerated expansion attributed to dark energy. Finally, we discuss some of the major open questions in modern cosmology, including the nature of dark matter and dark energy, the origin of cosmic inflation, and the current tensions between different cosmological observations.

This overview aims to provide a unified picture of our current understanding of the Universe while highlighting the fundamental questions that continue to drive research in cosmology.

Key words: *Cosmology; Big Bang; Cosmic Inflation; Dark Energy.*



Le paradoxe de l'âge du soleil l'épopée scientifique de la datation solaire

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Abstract

Déterminer l'âge du Soleil a constitué l'un des plus grands défis de l'histoire des sciences, provoquant au XIX^e siècle un vif débat entre la physique et la géologie. Alors que Lord Kelvin limitait l'âge solaire à 40 millions d'années par le calcul de la contraction gravitationnelle, les géologues et les darwiniens affirmaient que l'érosion des roches et l'évolution des espèces exigeaient des centaines de millions d'années. Ce paradoxe fut résolu au XX^e siècle par la découverte de la fusion nucléaire, révélant une source d'énergie colossale capable d'alimenter une étoile sur le très long terme. Ne pouvant dater directement les gaz solaires, la science se tourna vers la géochimie et les météorites primordiales, vestiges de la brique initiale de notre système. En 1953, Clair Patterson utilisa la méthode radioactive Uranium-Plomb pour fixer définitivement l'âge du Soleil à 4,57 milliards d'années. Pour réussir ses mesures, Patterson dut inventer le concept de chambre blanche, une quête de pureté qui mit en lumière la contamination mondiale de l'atmosphère par l'essence au plomb et mena à son interdiction historique. Aujourd'hui, l'héliosismologie ausculte les vibrations internes du Soleil et confirme que notre étoile est au milieu de sa vie. Son avenir est désormais tracé : dans un milliard d'années, sa luminosité croissante évaporerait les océans terrestres, avant qu'elle ne gonfle en Géante Rouge dans cinq milliards d'années, pour finir son existence en Naine Blanche au cœur d'une nébuleuse planétaire.

Keywords: *âge du Soleil, débat Kelvin-Helmholtz, Clair Patterson, datation Uranium-Plomb, météorites, héliosismologie, géante rouge.*



Oral Presentations



A Universe in Conflict: The Hubble Tension

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Abstract

The Hubble tension is one of the most significant unresolved problems in modern cosmology. It arises from a persistent discrepancy between the value of the Hubble constant (H_0) inferred from observations of the early universe, primarily through the cosmic microwave background within the standard Λ CDM cosmological model, and the value obtained from direct measurements of the local universe using distance indicators such as Cepheid variables and Type Ia supernovae. As observational precision has improved, this disagreement has grown increasingly difficult to attribute to statistical fluctuations or measurement errors, suggesting the possibility of unrecognized systematic effects or the need for new physics beyond the current cosmological framework. This presentation introduces the concept of the Hubble tension by reviewing the methods used to determine the expansion rate of the universe, comparing the principal observational techniques, and examining the evidence supporting the discrepancy. It also discusses the leading hypotheses proposed to resolve the tension, including improved calibration of astronomical distance measurements, modifications to the standard cosmological model, and extensions involving new particles or forms of dark energy. Finally, the presentation highlights the implications of the Hubble tension for our understanding of cosmic evolution and the future directions of observational and theoretical cosmology.



Cross-catalogue Classification of TESS Variable Stars with Random Forest

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Abstract

Machine-learning (ML) techniques have become increasingly important for the automated classification of variable stars in large photometric surveys. However, different classifiers are often trained using different reference catalogues, feature sets, and classification schemes, raising the question of how consistent their classifications are when applied to the same data. In this work, we apply a Random Forest classifier trained on variable stars from the General Catalogue of Variable Stars (GCVS) to a catalogue of variable stars identified from TESS observations by Wang et al. 2025. The classifier is trained using light curve features extracted from TESS photometry together with the Gaia G magnitude and the ($G_{BP}-G_{RP}$) color index, and covers a broad range of variability classes, including pulsating, eclipsing, and rotational. Evaluated on an independent validation set drawn from the GCVS, the model achieves a recall of 0.86 and a precision of 0.92. We then apply the trained classifier to the Wang et al. 2025 catalogue and compare the resulting classifications with those reported by the authors in order to investigate the level of agreement between two independent ML approaches based on different training catalogues. Overall, the classifications show a high degree of consistency (71%). The largest discrepancies are found for W Ursae Majoris-type eclipsing binaries (EW), which are most frequently classified by our model as rotatiting or δ Scuti variables. In fact, their light curves suggest that these discrepancies arise from the morphological similarity of low-amplitude contact binaries to rotational modulation or short-period pulsations in TESS photometry, as well as from differences in the adopted training catalogues. This study shows the transferability of supervised ML classifiers across independently constructed variable star catalogue.

Keywords: *Variable Stars, Catalogue, Machine Learning.*



Beyond Technical Obstacles: A Structural-Ontological Diagnosis of the Quantum Gravity Problem

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Abstract

Modern physics rests upon three highly succesful theoretical frameworks—quantum mechanics (QM), general relativity (GR), and quantum field theory (QFT)—whose empirical records are individually beyond dispute. However ,Despite this success, the standard formalizations of these theories employ mathematical structures that inherently resist unification. This structural fragmentation occurs directly at the level of the mathematical objects each theory uses to represent physical states, which reveals the blatant theoretical tensions such as between linear Hilbert spaces and non-linear solution manifolds, or between external time parameters versus dynamical geometry, fixed backgrounds versus background independence, non- commutative operator algebras versus commutative Poisson algebras To systematically categorize these key tensions, this paper introduces a precise diagnostic instrument: the structured state space (S, Σ) . This tool defines a physical system via a state space S equipped with a signature Σ that comprehensively encodes its algebraic, geometric, symmetry, temporal, observable, and compositional structures. We deploy this tool to perform a precise, dimension-by-dimension comparison of QM, GR, and QFT, generating a comprehensive catalog of their structural mismatches. We then apply this diagnostic lens to the existing proposed resolutions—including string theory, the AdS/CFT correspondence, loop quantum gravity, asymptotic safety, and causal set theory—to assess exactly which structural dimensions are successfully addressed and which persistent tensions remain. This exercise is explicitly diagnostic rather than foundational; it analyzes specific formalizations without claiming these structures are ontologically necessary features of physical reality. By mapping the exact structural vertices where existing unification strategies fail to converge, this diagnosis extracts six necessary conditions (R1–R6) required for any genuinely unified physical framework. Thus

,Ultimately, this structural taxonomy establishes the rigid constraints necessary to transition from the diagnostic role of this paper to the constructive, theoretical agenda for the future foundational work .

Keywords: *Quantum Gravity, Structural Incompatibility, structured state-space, Representation-Independent Description. .*



Hairy compact objects in Degenerate-Higher-Order-Scalar-Tensor (DHOST) theories

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Abstract

This work is devoted to investigating hairy compact object solutions within the framework of Degenerate Higher-Order Scalar-Tensor Theories, focusing on the interplay between scalar degrees of freedom and strong gravitational fields. These theories constitute the most general class of scalar-tensor models free from the Ostrogradsky instability despite the presence of higher-order derivatives, and provide a viable extension of General Relativity compatible with gravitational-wave constraints. We construct static and spherically symmetric compact configurations endowed with nontrivial scalar hair and analyze their physical and geometrical properties. Particular attention is devoted to the effects of the scalar sector on the interior structure, compactness, stability conditions, and spacetime geometry of neutron stars and black-hole-like compact objects. By solving the modified field equations numerically for several choices of the DHOST coupling functions and realistic equations of state, we show that the scalar hair can significantly modify the mass-radius relation, the pressure and density profiles, and the effective propagation of perturbations. Our results demonstrate that DHOST theories admit physically viable hairy compact objects and provide a rich phenomenology that may help constrain modified gravity in the strong-field regime.

Key words *scalar tensor theories, neutron stars, black holes.*



Variations in solar differential rotation during solar cycles 23–25

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Abstract

Solar rotation is one of the most fundamental properties of our nearest star. It drives and sustains the solar magnetic cycle through a large-scale dynamo operating from the base of the convection zone up to the photosphere. It was among the earliest observed solar phenomena, first noted by Galileo Galilei approximately 400 years ago, and remains a critical quantity to measure due to its temporal and spatial variability. Thanks to helioseismology, the solar rotation profile is now known not only at the surface but also in the deeper interior. It is established that rotation varies with latitude throughout the convection zone and exhibits temporal variations of a few m/s, known as torsional oscillations, which are closely related to the solar magnetic cycle. Using a local helioseismology technique called ring-diagram analysis, we refine our view of solar rotation by measuring its radial gradient in the first few megameters below the surface, where a shear layer is believed to play a central role in the generation and maintenance of the solar cycle. A butterfly-like diagram is revealed in this radial gradient, which appears to support the hypothesis of a near-surface seat of the solar magnetic field.

Keywords: Sun, helioseismology, solar rotation, solar activity, solar cycle



Poster Presentations



Stellar s-Process Nucleosynthesis: (n,γ) Reaction in the keV Range

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Abstract

The slow neutron capture process (s-process) plays a fundamental role in the stellar production of heavy elements in the universe. In this work, we propose to investigate the (n,γ) reaction at reference energies relevant to stellar environments. The study will focus on calculating the energy dependent cross section $\sigma(E)$ to illustrate the variation in capture probabilities. Furthermore, the Maxwellian-Averaged Cross Section (MACS) will be evaluated as a function of neutron energy, providing insight into how capture probabilities evolve with stellar temperature. From these results, the nuclear capture rate be determined, along with the characteristic timescale. Together, these quantities will offer a comprehensive picture of neutron capture dynamics in the s-process, contributing to a deeper understanding of nucleosynthesis pathways and the formation of heavy elements in astrophysical environments.

Keywords: s-process, neutron capture, MACS, nucleosynthesis, stellar physics



Photometric modeling of a Short-Period Eclipsing Binary TIC 273854760

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Abstract

The Hubble tension is one of the most significant unresolved problems in modern cosmology. It arises from a persistent discrepancy between the value of the Hubble constant (H_0) inferred from observations of the early universe, primarily through the cosmic microwave background within the standard Λ CDM cosmological model, and the value obtained from direct measurements of the local universe using distance indicators such as Cepheid variables and Type Ia supernovae. As observational precision has improved, this disagreement has grown increasingly difficult to attribute to statistical fluctuations or measurement errors, suggesting the possibility of unrecognized systematic effects or the need for new physics beyond the current cosmological framework. This presentation introduces the concept of the Hubble tension by reviewing the methods used to determine the expansion rate of the universe, comparing the principal observational techniques, and examining the evidence supporting the discrepancy. It also discusses the leading hypotheses proposed to resolve the tension, including improved calibration of astronomical distance measurements, modifications to the standard cosmological model, and extensions involving new particles or forms of dark energy. Finally, the presentation highlights the implications of the Hubble tension for our understanding of cosmic evolution and the future directions of observational and theoretical cosmology.



Evolution of the Sun's Internal Structure: Radial Profiles Across Key Evolutionary Stages

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Abstract

Stars undergo profound internal transformations throughout their lives, as nuclear burning processes reshape their density, temperature, and pressure structure. Using the Sun as a case study, this work investigates the evolution of its internal physical profiles across three key evolutionary stages: the early pre-main sequence (PMS), the zero-age main sequence (ZAMS), and the terminal-age main sequence (TAMS). Using the MESA (Modules for Experiments in Stellar Astrophysics) stellar evolution code, we compute and compare the radial profiles of density, temperature, and pressure at each stage, tracing how the Sun's internal structure adjusts as it contracts, ignites hydrogen fusion, and gradually exhausts its core fuel. The resulting profiles highlight the physical mechanisms driving the Sun's structural change, offering an intuitive visualization of its life cycle from formation to the end of core hydrogen burning. This approach provides both a pedagogical and quantitative tool for understanding solar evolution, illustrating how fundamental physics governs the transformation of the Sun over time.

Keywords: *solar evolution, stellar internal structure, MESA code, pre-main sequence, zero-age main sequence, terminal-age main sequence*



MCTDH Wave-Packet Study of Proton–Hydrogen Charge Transfer Dynamics for Energetic Neutral Atom Production in the Heliosphere

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Abstract

Charge transfer in proton–hydrogen collisions, $H^+ + H(1s) \rightarrow H(1s) + H^+$, is a resonant, symmetric process that governs the interaction between the solar wind and the local interstellar medium at the heliospheric boundary. Through this reaction, energetic solar wind protons are neutralized into Energetic Neutral Atoms (ENAs), which travel along straight trajectories and are directly measured by missions such as IBEX and Cassini/INCA, offering a unique remote-sensing probe of the heliosheath and of the heliopause location. Global heliosphere models still rely mainly on semi-empirical charge-transfer cross sections, whose accuracy remains uncertain at low collision energies, which directly affects the inferred size and structure of the heliosphere.

In this work, we investigate the quantum dynamics of the $H^+ - H$ charge-transfer reaction using the Multiconfiguration Time-Dependent Hartree (MCTDH) method, propagating nuclear wave packets on the coupled electronic states of the H_2^+ quasi-molecule. This time-dependent treatment gives access to the energy dependence of the charge-transfer cross section, together with state-to-state transition probabilities and interference (Stueckelberg-type) oscillation patterns arising from the coupled-channel dynamics, over an energy range relevant to the heliospheric interface. The resulting cross sections are intended to provide refined input data for global heliosphere and ENA-imaging models.

Keywords: Charge transfer, MCTDH quantum dynamics, Energetic Neutral Atoms, Heliosphere.



Nuclear Reactions of Astrophysical Interest: γ -ray Production Cross Sections for Solar Flare Modeling and Nuclear Astrophysics

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Abstract

Nuclear reactions occurring under the extreme conditions of temperature, density, and magnetic energy release play a central role in shaping the physical processes that govern our Sun and the stars. In stellar interiors, charged-particle induced reactions between light nuclei power hydrogen and helium burning and drive the nucleosynthesis of the elements; their cross sections and reaction rates, most often measured at astrophysically relevant sub-Coulomb energies, remain among the most important inputs to stellar evolution models. Beyond the quiescent stellar interior, nuclear reactions also manifest impulsively at the Sun's surface: during solar flares, protons and heavier ions accelerated to high energies by magnetic reconnection collide with ambient nuclei of the solar atmosphere (e.g. ^{12}C , ^{16}O , ^{24}Mg), producing characteristic de-excitation gamma-ray lines that provide a diagnostic of the accelerated particle population, composition, and energy content of these eruptive events.

This talk will review these two facets of nuclear astrophysics before presenting our own contributions to the field. We report γ -ray production cross section measurements for proton induced reactions on nuclei abundant in the solar atmosphere and the interstellar medium (e.g. ^{12}C , ^{16}O , ^{24}Mg , ^{28}Si , ^{56}Fe), performed at the Orsay tandem accelerator and at the SSC facility of iThemba LABS over incident energies extending up to 200 MeV. These data, compared to TALYS and EMPIRE-II nuclear reaction code predictions, are used to improve existing γ -ray line production cross section databases for modeling solar flare emission.

Keywords: nuclear astrophysics, reaction cross sections, solar flare.



Dark Matter Dark Energy

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Abstract

My educational poster presents a simplified journey through two of the greatest mysteries in modern cosmology: dark matter and dark energy. It begins with an overview of the four fundamental interactions, with particular emphasis on gravity and Einstein's revolutionary interpretation of it as the warping of the fabric of spacetime. The poster then explains planetary motion before introducing the Andromeda Galaxy, where the unexpectedly high orbital velocities of its outer stars led to the proposal of dark matter to account for the missing gravitational effects. It further illustrates how light serves as our primary source of information about the Universe, explaining the Doppler effect, redshift, and Hubble's discovery of the expanding Universe. Finally, it discusses the unexpected acceleration of cosmic expansion and the dark energy hypothesis proposed to explain this phenomenon.

Rich illustrations, explanatory diagrams, and QR codes linking to interactive simulations are included to enhance understanding and encourage further exploration of modern cosmology



Constraining the Generalized Uncertainty Principle via Gravitational Light Deflection

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Abstract

In this work, we have explored the theoretical and phenomenological consequences of a minimal length scale, introduced via a generalized uncertainty principle (GUP), on Hawking radiation. We have derived the corresponding entropy correction, obtaining a logarithmic term in agreement with various approaches to quantum gravity. This led us to reinterpret the temperature correction as arising from a modification of the Schwarzschild geometry, with an effective metric. Interestingly, this deformation affects not only black hole thermodynamics, but also observables accessible to classical tests, such as light deflection. In the weak field regime, we have derived the GUP correction to the solar deflection angle and obtained upper bounds on the deformation parameter. Our results, suggest that GUP-inspired deformations can leave imprints in both the strong- and weak-gravity regimes. Nevertheless, it is shown that classical gravitational tests cannot probe with high precision the quantum effect of gravity at macroscopic scales.

Keywords: *Minimal length, GUP, Hawking radiation, Light deflection.*



Investigating The Worlds Beyond the Dust - The case study of Stephenson 2-18 -

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Abstract

Stephenson 2-18, a red supergiant, is one of the largest stars currently known, and represents an extreme case of late-stage stellar evolution. This poster presents a review of its physical properties, including its exceptional radius, luminosity, and mass-loss rate in the context of current models of massive star evolution. Such extreme characteristics suggest that the red supergiant phase is not yet fully understood, and may involve more complex processes than standard models predict. The implications of these environments are further considered for planetary system formation. Strong radiation fields and intense stellar winds are expected to hinder the formation and long-term stability of exoplanets. However, studies of evolved stars, particularly red giants, indicate that planetary systems can exist and evolve under certain conditions, although their applicability to red supergiants remains uncertain. These considerations highlight current limitations in our understanding of both stellar evolution and planetary system survival in extreme environments.

Keywords: Red supergiants, stellar evolution, exoplanets, detection.



A Multi-Index Analysis of Geomagnetic Storms in Solar Cycle 24

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Abstract

This study presents a multi-index statistical analysis of geomagnetic storms during Solar Cycle 24 (2009–2019). By combining the *Dst* and Mayaud *aa* indices, geomagnetic storms were identified and classified according to their intensity, commencement type and rank. The analysis of the *Dst* index classifies the storms as moderate ($-100 \text{ nT} < Dst \leq -50 \text{ nT}$) or intense ($Dst \leq -100 \text{ nT}$). On the other hand, by analyzing the *aa* index, the storms are further classified according to their initial impulse into sudden-commencement storms (S-storms) or gradual commencement storms (G-storms). The analysis of their interplanetary drivers reveals clear differences, where: 76% of S-storms were driven by ICMEs and shock/sheath structures, particularly during the increasing and maximum phases of the solar cycle, 60% of G-storms were driven by CIRs associated with coronal holes, especially during the decreasing phase of the solar cycle. A long-term analysis covering Solar Cycles 11 to 24 (1868–2021) reveals a strong correlation between the sunspot number (SSN) and S-storm occurrence ($R = 0.90$), while the correlation with G-storm occurrence is considerably weaker ($R = 0.30$). These results highlight the distinct solar origins for different storm types and demonstrate the value of multi index classification for characterizing geomagnetic storm occurrence and evolution.

Keywords: Solar cycle 24, Geomagnetic storms, *Dst* index, *aa* index.



X-ray Spectral Analysis of an AGN: Reflection and Variability in Mrk 841

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Abstract

This study presents an X-ray spectral analysis of the Seyfert 1 galaxy Mrk 841 ($z = 0.0365$) using four archival XMM-Newton EPIC-pn observations: three from January 2001 and one from January 2022. The analysis band is 3.5–10 keV, covering the hard power-law continuum. The soft X-ray excess below 3.5 keV is excluded in this work because of the strong soft excess. We first fit the observations with an absorbed power-law, we then add gaussians to fit the emission lines, and finally, depending on phenomenological fitting results we use physical models of relativistic and non relativistic reflection. In order to study the variability of our target over the time we test tying some parameters during the fitting. Physical models from the relxill package are applied in the third stage of fitting : xillver [2] for Obs. I, II, and IV, and relxill for Obs. III. All spectra use χ^2 as the fit statistic and are grouped to a minimum of 20 counts per bin. Galactic absorption is ($N_H = 0.02 \times 10^{22} \text{ cm}^{-2}$) and source redshift is ($z = 0.0365$). The results are compared with previous studies of Mrk 841, 2001's analysis by Longinotti et al. and Petrucci et al. and 2022 analysis by Mehdipour et al. The work aims to investigate the temporal variability of Mrk 841 and to constrain its fundamental physical properties, including the ionisation state of the accretion disc and its geometry.

Key words: *active galactic nuclei , X-ray spectral analysis , iron emission line reflection , spectral variability , Mrk 841*



mKdV Solitary Waves and Double-Layer Structures in Quark-Gluon Plasmas: A Parametric Study

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Abstract

The modified Korteweg-de Vries (mKdV) equation is a fundamental nonlinear framework used to model localized wave phenomena in complex media, admitting both solitary wave and double-layer solutions. The emergence and propagation of these structures are strictly governed by the sign of the quadratic nonlinearity coefficient, α_2 , with solitary waves forming when $\alpha_2 > 0$ and double layers appearing when $\alpha_2 < 0$. This study presents a parametric numerical investigation into the behavior of α_2 within a quark-gluon plasma, focusing on the critical roles played by the equilibrium density (ρ_0), the coupling constant (g), and the hard gluon dynamic mass (m_g). By mapping the sign of α_2 across these physical parameters, we delineate the exact existence regions and transition boundaries for both solitons and double layers. The results demonstrate that variations in density, coupling strength, and gluon mass drastically alter the conditions for wave propagation, providing key insights into the nonlinear dynamics of hot, dense nuclear matter.

Keywords: Relativistic Hydrodynamics, mKdV equation, Solitons, Double- Layer Structures .



General Relativity Within the Vierbein Formalism in Astrophysics

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Abstract

The vierbein (tetrad) formalism offers a geometrically transparent framework for the description of gravitation based on local Lorentz invariance. It replaces the metric tensor with a set of orthonormal basis fields defined at each spacetime point, allowing a natural coupling to spinorial matter and a gauge- theoretic interpretation of gravity. In this context, General Relativity is reformulated using tetrads and spin connections, leading to field equations fully equivalent to Einstein's equations while providing deeper insight into the roles of curvature and local symmetries. The variational structure of the Einstein–Hilbert action is also revisited within this formalism. Finally, the tetrad approach is applied to the Schwarzschild spacetime, where an explicit vierbein construction is presented and used to analyze the local geometry surrounding a spherically symmetric compact object. This application illustrates how the vierbein formalism provides a convenient framework for describing gravitational phenomena in astrophysical environments and serves as a foundation for studying particle dynamics and local inertial effects in strong gravitational fields

Keywords: *General Relativity, Torsion, Vierbein, Schwarzschild spacetime.*



Application of the Generalized Uncertainty Principle to Black Hole Remnants.

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Abstract

One of the most important open questions in theoretical physics is the evaporation of black holes. Small black holes are expected to emit blackbody radiation at the Hawking temperature until they approach the Planck scale, after which their evolution remains highly uncertain. A commonly held argument suggests that the absence of a conserved quantum number causes black holes to evaporate completely into photons and other ordinary stable particles and vacuum, in a manner similar to any unstable quantum system.

In order to address this open problem, In this work we apply a new expression of the Generalized Uncertainty Principle (GUP) to black hole thermodynamics. The GUP expression employed here is derived within the framework of the generalized Gross-Pitaevskii equation for both small and large self-bound quantum droplets. We calculate the GUP-corrected black hole temperature and the GUP-corrected black hole entropy. The expected results suggest that the GUP enforces a minimum mass and size for evaporating black holes, preventing complete evaporation and leading to the formation of a stable black hole remnant. Notably, having a finite temperature object as an end state of Hawking evaporation is arguably more physically reasonable than the divergent temperature predicted by the standard picture of Hawking evaporation

Keywords: Generalized Uncertainty Principle, black hole remnants, black hole temperature, self-bound quantum droplets.



The Energy Spectrum of The Hydrogen Atom in The Presence of a Minimal Length

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Abstract

During recent years, there has been much interest in studying theories characterized by a minimal observable length. Some works in the context of perturbative string theory have indeed suggested a generalized uncertainty principle. Several problems related to minimal length have been studied, including the harmonic oscillator, the hydrogen atom, and the Casimir effect. In this work, we calculate the correction to the energy spectrum of the hydrogen atom arising from the presence of a minimal length. We introduce the minimal length scenario by considering a noncommutative model in the presence of a minimal length. The concept of minimal length has been incorporated into the theory through the extension of Heisenberg's algebra, as selected by Kempf, and in the specific instance where $\beta'=2\beta$. We calculate the energy spectrum of the hydrogen atom in first order of the parameter related to the minimal length via perturbation theory by introducing the Coulomb potential in the new Dirac energy operator.

Keywords: hydrogen atom, Dirac equation , minimal length,.



Calculation of Charge Exchange Cross Sections between Solar Wind Protons and Neutral Hydrogen in the Martian Exosphere over the 0–2 keV Energy Range

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Abstract

This work presents a physical study of the charge exchange process occurring between energetic solar wind protons and neutral hydrogen atoms in the Martian exosphere, over the energy range of 0–2 keV. When a solar wind proton captures an electron from a neutral hydrogen atom, it is converted into an Energetic Neutral Atom (ENA), while the neutral hydrogen atom becomes a proton. This process plays a fundamental role in the formation of the population of energetic neutral atoms that subsequently penetrate the Martian atmosphere. The objective of this study is to calculate the charge exchange cross sections as a function of proton energy over the 0–2 keV range, in order to characterize this mechanism as a key initial step in the generation of proton aurora on Mars

Keywords: *Charge Exchange, Solar Wind, Exospheric Hydrogen, Cross Sections .*



Calculation of Radiative Excitation Cross Sections Induced by Energetic Neutral Atoms in the Martian Atmosphere

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Abstract

This work presents a physical study of the radiative excitation process resulting from collisions between Energetic Neutral Atoms (ENAs) of hydrogen and neutral atmospheric molecules, such as carbon dioxide (CO₂), in the atmosphere of Mars. As the hydrogen ENAs penetrate the Martian atmosphere, they collide with CO₂ molecules, transferring part of their kinetic energy and promoting the hydrogen atoms to higher electronic states. The subsequent spontaneous decay of the excited hydrogen atoms to the ground state produces the characteristic photon emission (Lyman-alpha, 121.6 nm) observed in proton aurora. The objective of this study is to calculate the radiative excitation cross sections associated with these H-CO₂ collisions, as a function of ENA energy, given their essential role in producing the characteristic radiative emission of proton aurora observed in the atmosphere of Mars

Keywords: Radiative Excitation, Energetic Neutral Atoms (ENAs), Carbon Dioxide, Cross Sections .



Spectral Engineering for Space-Grade Photovoltaics: A DFT+U_d+U_p Study of Lanthanide-Doped Rutile TiO₂

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Abstract

In modern space exploration, lanthanides (rare-earth elements) have emerged as indispensable building blocks, driving critical technologies from high-temperature Nd/Sm-based magnets in ion thrusters to Erbium-doped fiber amplifiers in deep- space laser communications. Extending their unique optical properties to energy harvesting, lanthanide-doped materials offer a revolutionary pathway to bypass the theoretical efficiency limits of space photovoltaics through spectral conversion. While rutile TiO₂ is a premier candidate for space applications due to its exceptional tolerance to cosmic ionizing radiation and thermal stability in vacuum, its wide bandgap severely limits solar light harvesting.

To address this challenge, we present a rigorous first-principles investigation of rutile doping with selected Lanthanides (Ln = Er, Nd) designed to induce up- conversion (IR-to-visible) and down-conversion (UV-to-visible) mechanisms. Calculations were performed using a representative 2x2x3 supercell (72 atoms), a model optimized to prevent artificial periodic interactions while ensuring computational feasibility [1]. To overcome the self-interaction errors of standard DFT, a highly accurate DFT+U_d+U_p formalism was employed to correctly resolve the strongly correlated 4f states of the lanthanides, 3d of titanium, and 2p of oxygen. Our results reveal that lanthanide integration introduces discrete, well-positioned intermediate bands within the rutile TiO₂ bandgap. This electronic restructuring significantly broadens the optical absorption spectrum into the visible and infrared ranges while promoting efficient electron-hole separation [2]. This work provides critical atomistic insights for engineering next-generation, radiation-hardened, and highly efficient solar cells for long-duration space missions.

Keywords: Rutile TiO₂, Lanthanide doping, 2x2x3 supercell, DFT+U_d+U_p, Space-grade photovoltaics, Spectral conversion (up/down-conversion).

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Optimization of Multilayer Shielding for Galactic Cosmic Ray Cancer Risk Mitigation in Long- Duration Deep-Space Missions

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Abstract

Deep space exploration beyond the terrestrial magnetosphere presents unprecedented challenges that critically threaten crew survivability and overall mission efficacy, chief among which is the hazard of Galactic cosmic ray (GCR) exposure, which elevates carcinogenesis risks during long-duration spaceflight. This study provides a comparative analysis of multilayer shielding material configurations designed to reduce GCR- induced cancer risk through optimized radiation attenuation. Using NASA's Online Tool for the Assessment of Radiation In Space (OLTARIS), we evaluated three distinct shielding approaches: conventional water shielding, aluminium-polyethylene-boron-lithium (ABL) composites, and aluminium- polyethylene-graphite-epoxy (APG) systems across a standardized mission profile of 356 days. The results demonstrate that advanced multilayer composite materials significantly outperform traditional single-material shielding, with ABL configurations achieving an 89.6% reduction in total cancer risk (from 13.54% unshielded baseline to 1.40% with optimization) compared to 85.8% risk reduction for APG systems. The superior performance of ABL materials is attributed to enhanced neutron attenuation via boron and lithium constituents, which effectively mitigate secondary radiation effects and consider as a critical factor often overlooked in conventional protection strategies. The methodology and findings herein establish a framework for continued optimization of radiation protection strategies essential to sustainable long-duration human space exploration.

Keywords: Deep space radiation, Multilayer shielding, OLTARIS, Carcinogenesis risks.



NIRISS Instrument in Wide Field Slitless Spectroscopy mode: An Overview

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Abstract

In this article we cover how *Wide Field Slitless Spectroscopy* mode (**WFSS**)¹ data can provide us with information about redshifted galaxies in the early universe, using the *Near-InfraRed Imager & Slitless Spectrograph* (**NIRISS**)² Instrument aboard the *James Webb Space Telescope* (**JWST**)³, and its most crucial methods and components. operating between wavelengths of $0.8\mu\text{m}$ - $2.2\mu\text{m}$

Keywords: 1.WFSS, 2.NIRISS, 3.JWST



Monte Carlo Optimization of a Backscattering Device for Planetary Subsurface Exploration

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Abstract

Planetary exploration heavily relies on nuclear planetary science to detect water-equivalent hydrogen and ice in subsurfaces, as successfully demonstrated by planetary missions like NASA's Lunar Prospector or the DAN instrument aboard the Curiosity rover. These missions utilize neutron spectrometry to measure the signature of particles interacting with planetary soil. Inspired by these technologies, this work focuses on the numerical modeling and optimization of an experimental neutron backscattering device using the MCNP (Monte Carlo N-Particle) transport code. The main objective is to enhance the system's sensitivity and spatial resolution by optimizing its geometric configuration, shielding materials, and detector placement. Through extensive stochastic simulations, we analyzed the influence of neutron moderation, scattering cross-sections, and background radiation on the backscattered signal. The optimized design significantly improves the signal-to-noise ratio, demonstrating high accuracy in characterizing varying soil composition and identifying hydrogenous materials behind structural barriers. These findings provide a preliminary study for the development of robust instrumentation payloads, demonstrating how advanced numerical modeling can support the emergence of local scientific research initiatives for both specific terrestrial and planetary applications.

Keywords: MCNP Optimization, Backscattering, Space Techniques, Subsurface Exploration.



The Instability of Kelvin–Helmholtz in the Astrophysical Jets

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Abstract

The Kelvin–Helmholtz instability (KHI) arises at the interface between two fluids moving at different velocities, where velocity shear amplifies small perturbations into vortex-like structures and turbulent mixing. This work examines the role of KHI in astrophysical jets — highly collimated, high-velocity plasma outflows launched from compact objects such as black holes, neutron stars, and young stellar objects. We base our analysis on observational results reported by Li, Zhang, Yang, Hou, and Erdélyi, which provide the first direct observation of KHI in a solar blowout jet using high-resolution IRIS 1400 Å data. The study tracks the jet boundary evolving from a smooth profile to a sawtooth pattern, allowing estimation of shear velocities ($\sim 204 - 264$ km/s) and vortex scales (~ 1.6 Mm) consistent with KHI growth. The instability condition, $w^2 = -k^2[r_1r_2/(r_1+r_2)^2](\Delta v)^2$, shows that perturbations grow exponentially when $w^2 < 0$. Extending this framework to astrophysical jets in general, KHI at jet boundaries or internal shear layers destabilizes jet structure, drives mixing with the surrounding medium, and converts bulk flow energy into heat and particle acceleration. Studying KHI in solar jets thus provides an observational testbed for the same physics governing much larger jets from black holes and young stars, linking near-Earth space weather to the energy transport processes that shape galactic environments.

Keywords: *Kelvin–Helmholtz instability, astrophysical jets, solar blowout jets, plasma shear, IRIS*



Analysis of CME-Induced Perturbations in the Ionospheric D Region Using VLF Remote Sensing

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Abstract

Coronal Mass Ejections (CMEs) are among the most important solar transient events that can significantly affect the near-Earth space environment and ionospheric conditions. When a CME interacts with the Earth's magnetosphere, it can produce variations in the ionization level and conductivity of the ionospheric regions, particularly the D-region, which plays a crucial role in very low frequency (VLF) wave propagation. This work focuses on the analysis of CME induced perturbations in the ionospheric D-region using VLF remote sensing techniques. Variations in VLF signal parameters, such as amplitude and phase, are investigated as indicators of changes occurring in the lower ionosphere during solar disturbances. The sensitivity of VLF waves to ionospheric conductivity variations makes them an effective tool for monitoring space weather effects. The study aims to contribute to a better understanding of the response of the D-region ionosphere to CME events and to highlight the potential of VLF observations for the characterization and monitoring of solar-induced ionospheric disturbances.

Keywords: *Coronal Mass Ejection, Space Weather, Ionospheric D-region, VLF waves, Remote Sensing.*



Europa and Enceladus: Could They Be More Habitable Than Mars?

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Abstract

The search for habitable environments beyond Earth is one of the major goals of planetary science and astrobiology. Although Mars has traditionally been the primary target in the search for extraterrestrial life, recent discoveries suggest that the icy moons Europa and Enceladus may offer more favorable conditions for habitability. Both moons are believed to harbor global subsurface oceans beneath their icy crusts, where liquid water is maintained by tidal heating generated through gravitational interactions with Jupiter and Saturn, respectively.

Observations from previous space missions have also revealed evidence of water vapor plumes, organic molecules, and chemical compounds that could support prebiotic chemistry.

This poster reviews the physical and geological characteristics that make Europa and Enceladus promising candidates in the search for life beyond Earth. It discusses the essential requirements for habitability, including liquid water, energy sources, and the chemical building blocks of life, while comparing these ocean worlds with Mars. The poster also highlights the role of past, current, and future space missions in improving our understanding of these environments and their astrobiological potential.

The study emphasizes that icy ocean worlds have become key targets for future exploration and may significantly reshape the strategy for searching for life elsewhere in the Solar System.

Keywords: *Europa, Enceladus, Habitability, Astrobiology, Ocean Worlds, Solar System.*



Caractérisation du couplage atmosphère–ionosphère par une approche multi-instrument

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Abstract

Le couplage entre l'atmosphère neutre et l'ionosphère se manifeste lorsqu'une perturbation générée près du sol se propage verticalement et modifie, avec un certain délai, la densité électronique à différentes altitudes. Ce couplage reste difficile à démontrer directement, car il exige des observations simultanées à plusieurs niveaux de l'atmosphère, le long d'un même trajet de propagation.

L'éruption explosive du volcan sous-marin Hunga Tonga–Hunga Ha'apai, le 15 janvier 2022, l'un des événements géophysiques les plus énergétiques jamais enregistrés, a généré des ondes de Lamb et des ondes de gravité atmosphérique propagées à l'échelle planétaire. Le territoire algérien présente une particularité géographique déterminante : le point antipodal de l'éruption se situe dans le sud du pays. Les ondes traversent donc l'ensemble du territoire une première fois, convergent en ce point, puis repassent après réflexion, offrant une opportunité inédite d'observer l'aller et le retour du couplage vertical sur un même site.

L'objectif de ce travail est de mettre en évidence ce couplage au-dessus de l'Algérie à partir de trois observables complémentaires. Les mesures de pression atmosphérique, enregistrées simultanément à Alger et à deux stations britanniques (Honiton et Newchapel), montrent un même pic de pression décalée dans le temps entre les deux pays, correspondant à une vitesse de propagation d'environ 300 m/s, caractéristique d'une onde de Lamb ; deux passages supplémentaires sont ensuite identifiés, liés à la convergence puis au retour de l'onde. Dans la basse ionosphère, l'amplitude de signaux radio VLF/LF reçus à Alger présente des modulations cohérentes avec ces perturbations de pression. Dans la haute ionosphère, un réseau de 23 stations GNSS algériennes révèle des oscillations du contenu électronique total (TEC), de 0,4 à 0,7 TECU, avec une vitesse de propagation d'environ 283 m/s, compatible avec une onde de gravité atmosphérique et cohérente avec les résultats en pression.

La convergence des ondes au point antipodal se traduit, de façon synchrone dans les trois observables pression, VLF/LF et TEC GNSS, par une amplification marquée du signal, apportant une preuve directe du couplage vertical entre la surface et l'ionosphère. Cette étude constitue la première démonstration multi-instrumentale, en Algérie, de ce couplage pour un événement global.

Keywords: Ionosphere, Atmosphere, Tonga Hunga, Pression.



The Impact of Friedmann Equations on Cosmic Expansion: A Workshop

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Abstract

This workshop provides an in-depth examination of the role of the Friedmann equations in shaping our modern understanding of the universe's expansion. Derived from Einstein's field equations of general relativity under the assumptions of homogeneity and isotropy, the Friedmann equations form the mathematical and physical framework describing the time evolution of the cosmic scale factor. We will review their derivation and then discuss the influence of matter density, radiation, dark energy, and spatial curvature on expansion dynamics. The workshop focuses on linking these equations with contemporary cosmological observations, including Type Ia supernovae, the Cosmic Microwave Background (CMB), and Baryon Acoustic Oscillations (BAO), illustrating how they lead to the standard Λ CDM model. Participants will analyze different expansion scenarios (open, closed, flat universes) and understand the physical meaning of the Hubble parameter and deceleration factor, supported by simple numerical simulations. Finally, we address the limitations of these equations in light of current challenges such as the Hubble tension and the use of quasars as new standard candles.

Keywords: *Friedmann equations, cosmic expansion, general relativity, dark energy, Hubble constant, Λ CDM model.*